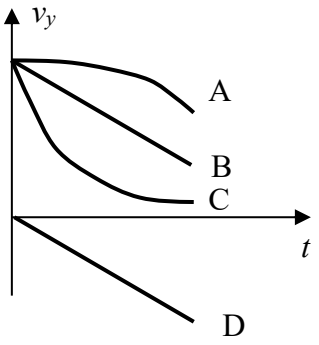
Fall 2026 **Physics 1135** Print LAST Name: solution

Rec Sec Number ____ **TEST 1 (4 pages)** and First Name: _____

For questions 1-5, write your answer in the underlined space **to the left of the question number**. For problems 6-8: begin with an Official Starting Equation. The final expression must be in system parameters and simplified as far as possible. Neglect air resistance. Calculators and notes cannot be used during the test. If you have questions, ask the proctor. **You must put your name on each page.**

Test Total =_____ / 200

_D____1. (10 points) Students drop a ball from the roof of the physics building. The v_y-t graph of the ball's motion is given by which plot letter in the graph at the right? The y-axis is directed upwards.



A) A B) B C) C D) D

_C____2. (10 points) The position of an object in one-dimensional motion as a function of time t is given by $y(t) = dt^3 + bt^2 - c$ where d, b, and c are positive constants. At time $t = 0$, the object's velocity and acceleration components are:

A) $v_y = 0$, and $a_y = 0$ B) $v_y = 2b$, and $a_y = 0$

C) $v_y = 0$, and $a_y = 2b$ D) $v_y = 2b$, and $a_y = 6d$

_C____3. (10 points) A ball is thrown with an initial speed at θ=30° above the horizontal. When it has reached **half** its maximum height, which quantity is the same as just after launch?

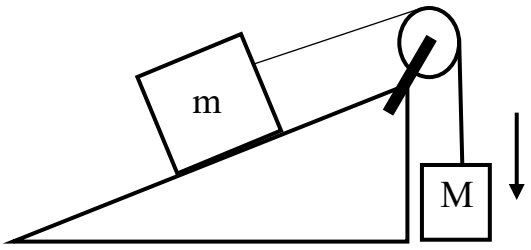
A) speed B) velocity C) horizontal component of velocity D) vertical component of velocity

_B____4. (10 points) Dr. Vojta's cat Raisin sleeps on a horizontal bookshelf. The reaction force to the weight force acting on Raisin is

A) the normal force on Raisin by the shelf. B) the force of gravity by Raisin on the Earth.

C) the force by Raisin on the shelf. D) There is no reaction force because Raisin is at rest.

_C____5. (10 points) Mass m on a frictionless incline is connected to mass M by a rope that runs over a massless, frictionless pulley. If mass M moves downwards at constant speed, the tension T in the rope is:



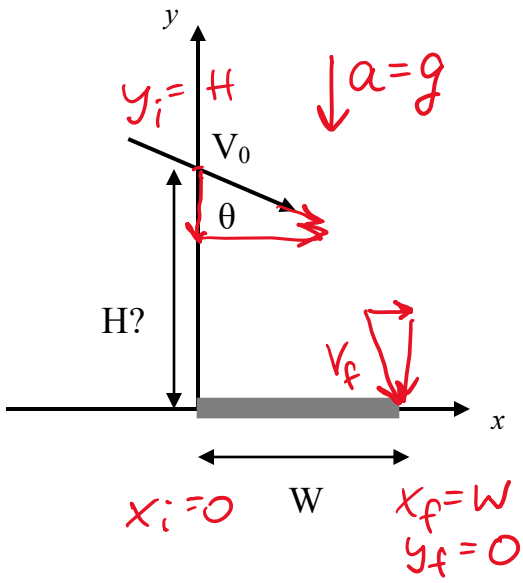
A) $T = (M - m)\, g$ B) $T < Mg$

C) $T = Mg$ D) $T > Mg$

__50___ / 50 for this page

Name:_____

6. (50 points) A foolish person executes a running leap off the roof of a building into a river below, just barely clearing a sidewalk of width *W* between the building and the river. The roof of the building slopes down, so that the person leaves the roof at **unknown** height *H* above the sidewalk, with initial speed V_0 at an angle θ with the **vertical**, as shown in the figure.



a) (10 points) Complete the diagram with all information needed to solve parts b and c below.

b) (20 points) Derive an expression for the height *H* above the sidewalk, in terms of system parameters.

$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$

$W = v_0 \sin\theta\, t$

$t = \frac{W}{v_0 \sin\theta}$

$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$

$0 = H - v_0 \cos\theta\, t + \frac{1}{2}(-g)t^2$

$H = v_0 \cos\theta \left(\frac{W}{v_0 \sin\theta}\right) + \frac{1}{2}g\left(\frac{W}{v_0 \sin\theta}\right)^2$

$H = W \cot\theta + \frac{1}{2}g\left(\frac{W}{v_0 \sin\theta}\right)^2$

c) (20 points) Derive an expression for the velocity with which the person hits the water, **in unit vector notation**, in terms of system parameters. Use *H* as a system parameter for this part.

$v_x = v_{0x} + a_x t$

$v_x = v_0 \sin\theta$

$v_y = v_{0y} + a_y t$

$v_0 = -v_0 \cos\theta - g\frac{W}{v_0 \sin\theta}$

OR

$v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$

$v_y^2 = (v_0 \cos\theta)^2 + 2(-g)(-H)$

$v_y = -\sqrt{(v_0 \cos\theta)^2 + 2gH}$

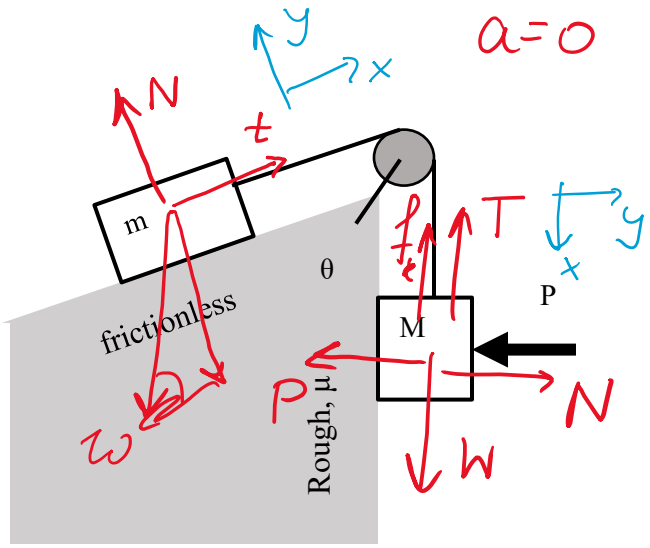
$\vec{v} = v_0 \sin\theta\, \hat{i} - \left(v_0 \cos\theta + \frac{gW}{v_0 \sin\theta}\right)\hat{j}$

OR $\vec{v} = v_0 \sin\theta\, \hat{i} - \sqrt{(v_0 \cos\theta)^2 + 2gH}\,\hat{j}$

50 / 50 for this page

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7. (50 points) A crate of mass *M* is lowered from the top of a ramp to the floor. The inclined ramp makes an angle θ with the **vertical**. A block of mass *m* moves along the **frictionless incline** and is connected to the box by a massless rope and a massless, frictionless pulley. The **vertical wall is rough** with a coefficient of kinetic friction μ between the wall and the box. A person pushes the crate against the wall with a **horizontal** force of magnitude *P*.



a) (10 points) In the diagram, superimpose fully labeled free-body diagrams for the box and the block. Add all information necessary to solve part b).

b) (40 points) Derive an expression for the force magnitude P for which the crate descends at constant speed, in terms of system parameters.

$m: \sum F_x = N_x + t_x + w_x = ma_x$

$t - mg\cos\theta = 0$

$T = mg\cos\theta$

$M: \sum F_x = T_x + f_{kx} + P_x + N_x + W_x = Ma_x$

$-T - f_k + Mg = 0$

$f_k = \mu N$

$\sum F_y = T_y + f_{ky} + P_y + N_y + W_y = Ma_y$

$N - P = 0$

$N = P$

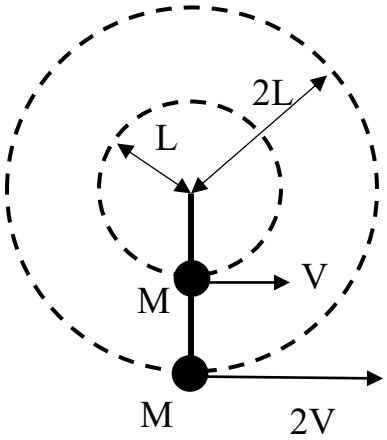
$-mg\cos\theta - \mu P + Mg = 0$

$$P = \frac{(M - m\cos\theta)\,g}{\mu}$$

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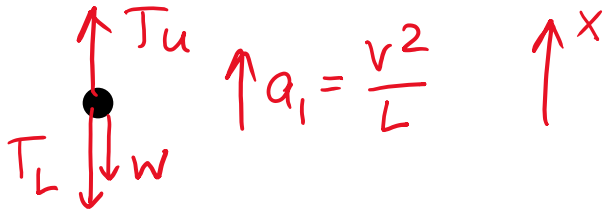
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8. Two identical balls are tied together, each of mass M, using a cord of length L. Another cord, also of length L, is tied to one of the balls and both balls are swung in a **vertical** circle. The figure shows the two balls at the instant when both are at their lowest points. At that instant, the speed of the upper ball is V, and the speed of the lower ball is $2V$.

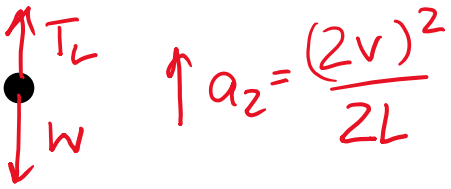


a) (10 points) Using the point masses drawn below, draw two fully-labeled free body diagrams, one for each ball ("upper" and "lower") and any other information needed to solve part b).

upper ball



lower ball



b) (40 points) Treating the balls as point masses, derive an expression for the tension in the upper cord at the instant shown in the figure, in terms of relevant system parameters.

Lower: $\Sigma F_x = T_{Lx} + W_x = M a_{2x}$

$$T_L - Mg = M\frac{(2v)^2}{2L} = M\frac{4v^2}{2L} = \frac{2Mv^2}{L}$$

$$T_L = Mg + 2\frac{Mv^2}{L}$$

Upper: $\Sigma F_x = T_{Ux} + T_{Lx} + W_x = Ma_{1x}$

$$T_U - T_L - Mg = M\frac{v^2}{L}$$

$$T_U = Mg + T_L + M\frac{v^2}{L}$$

$$\boxed{T_U = 2Mg + \frac{3Mv^2}{L}}$$

50 / 50 for this page